

Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

software Engineering [CSE-4042]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) You are assigned a software development project with evolving requirements and a high need for early testing. Apply your understanding of the Waterfall Model and its extensions to choose the most appropriate model for this scenario. Justify your choice by explaining how the selected model addresses the project's needs. (5)
 - A)
 - B) Your team is working on a project where customer feedback is expected to change frequently during development. Apply Agile development principles to explain how your team would handle these changes. Mention which specific Agile practices would be most beneficial in this scenario. (3)
 - C) The Spiral Model is drawn as a multi-loop spiral, with each loop representing one iteration of the development process. Draw and label the basic skeleton of the Spiral Model, highlighting its key phases. (2)
- 2) You are tasked with gathering and analyzing requirements for a new e-commerce website. Apply appropriate techniques to identify, document, and prioritize the needs of various stakeholders. Explain how you would ensure that the requirements are complete and aligned with business goals. (5)
 - A)
 - B) You have been assigned to create a Software Requirements Specification (SRS) document for an online food delivery application. Apply the key sections of an SRS document to organize the requirements effectively. Briefly explain how each section helps ensure the system meets stakeholder expectations. (3)
 - C) Identify the steps to apply user-centered design principles to improve the usability of a mobile application. (2)
- 3) Draw the sequence diagram for the use-case scenario "Create New Library User Account" by identifying appropriate synchronous and asynchronous messages. Analyze the object creation and deletion and fragments. (5)
 - A)
 - B) A company is developing an e-commerce platform where customers can browse products, place orders, and manage their accounts. Choose the important concepts and terminologies associated with a layered design for e-commerce platform. (3)
 - C) Develop a state diagram for a Ticket object in a ticketing system. Include states such as Available, Booked, Cancelled, and Closed. (2)

- 4) Draw a Level 1 DFD for a library management system, breaking down the key processes from the context diagram. Identify the main data stores, data flows, and processes that interact with external entities. (5)
- A)
- B) You are tasked with designing a library management system for a university. The system allows students and faculty to search for books, borrow them, and return them. Librarians are responsible for adding new books, managing overdue fines, and maintaining the catalog. Draw a Use Case Diagram for the library management system, and analyze their interactions with the system. (3)
- C) Consider an online shopping system where customers browse products and place orders.
Class: Product
Attributes: productID, name, price, stock.
Methods: updateStock(), applyDiscount().
Class: Order (2)
Attributes: orderID, orderDate, totalAmount.
Methods: calculateTotal(), addProduct().
Identify the role of objects and classes in the OOAD methodology for a given scenario.
- 5) You are tasked with testing a new e-commerce website. Apply the principles of functional testing to describe how you would ensure the system meets user requirements. Outline the steps you would take and the specific aspects of the website you would test. (5)
- A)
- B) You suspect there are memory leaks in a software application you're developing. Apply a program analysis tool to find and address memory-related issues in your code. (3)
- C) Apply the Singleton pattern to design a logging system for an application. How would this ensure only one instance of the logger exists? (2)

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